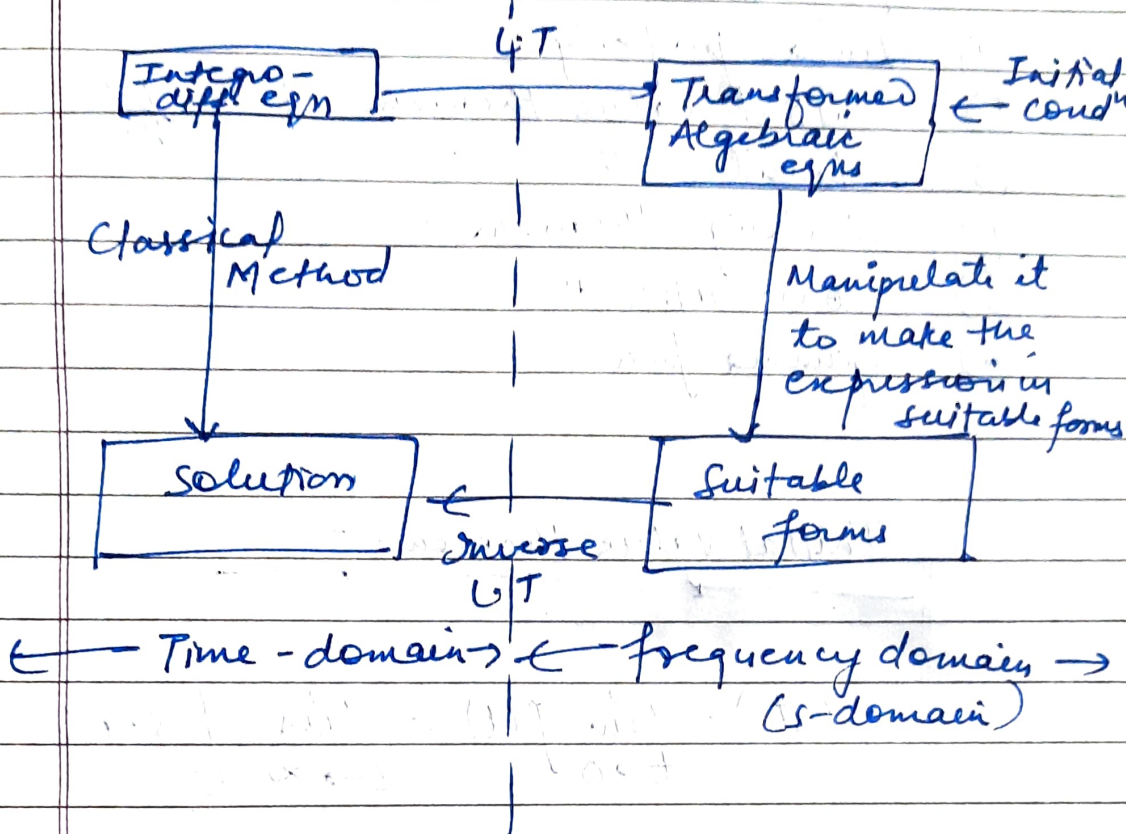


Laplace transform

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Definition of L.T

$$L[f(t)] = f(s)$$

$$f(s) = \int_0^{\infty} e^{-st} f(t) dt$$

where S = Laplace operator

$$S = \sigma + j\omega$$

$\downarrow$ Complex frequency $\downarrow$ Neper frequency $\rightarrow$ Radian/Angular frequency

Properties: —

- ① Linear property
- ② Initial value theorem
- ③ final " "
- ④ Scaling theorem
- ⑤ Convolution "
- ⑥ Gamma fun

Initial value theorem: —

$$f(0^+) = \lim_{t \rightarrow 0} f(t) = \lim_{s \rightarrow \infty} [s \cdot f(s)]$$

Final value theorem: —

$$f(\infty) = \lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} [s \cdot f(s)]$$

eg:- Determine the initial value $f(0^+)$ if

$$f(s) = \frac{2(s+1)}{s^2 + 2s + 5}$$

Soln:-

$$\begin{aligned} f(0^+) &= \lim_{s \rightarrow \infty} [s \cdot f(s)] \\ &= \lim_{s \rightarrow \infty} \left[\frac{2s(s+1)}{s^2 + 2s + 5} \right] \\ &= \lim_{s \rightarrow \infty} \left[\frac{2 + \frac{2}{s}}{1 + \frac{2}{s} + \frac{5}{s^2}} \right] = 2 \text{ Ans} \end{aligned}$$

eg 1 $f(0^+) & f(\infty) = ?$

(2) $f(s) = \frac{4e^{-2s}}{s} (s+50)$

(3) $f(s) = \frac{5s+3}{s(s+1)}$

(4) $f(s) = \frac{5s^3 - 1600}{s(s^3 + 18s^2 + 90s + 800)}$

* Laplace transform to waveforms

* Laplace transform to differential eqn

* Laplace transform to circuits

Laplace transform of some func:-

⑦

$f(t)$

$$f(s) = \int_0^{\infty} f(t)e^{-st} dt$$

① 1 or $U(t)$, k

$$\frac{1}{s}, \frac{k}{s}$$

② t, t^n

$$\frac{1}{s^2}, \frac{n!}{s^{n+1}}$$

③ $\delta(t)$

$$1$$

④ $e^{\pm at}$

$$\frac{1}{s \mp a}$$

⑤ $te^{\pm at}$

$$\frac{1}{(s \mp a)^2}$$

⑥ $\sin \omega t$

$$\frac{\omega}{s^2 + \omega^2}$$

⑦ $\cos \omega t$

$$\frac{s}{s^2 + \omega^2}$$

⑧ $e^{-at} \sin \omega t$

$$\frac{\omega}{(s+a)^2 + \omega^2}$$

⑨ $e^{-at} \cos \omega t$

$$\frac{s(s+a)}{(s+a)^2 + \omega^2}$$

⑩ $\sinh at$

$$\frac{a}{s^2 - a^2}$$

⑪ $\cosh at$

$$\frac{s}{s^2 - a^2}$$

⑫ $e^{\pm at} f(t)$

$$f(s \mp a)$$

⑬ $f(t \pm t_0)$

$$e^{\pm t_0 s} f(s)$$

Laplace transform of Signals :-

2

① Unit step fun:-

$$f(t) = U(t)$$

$$f(s) = \int_0^{\infty} e^{-st} f(t) dt = \int_0^{\infty} U(t) \cdot e^{-st} dt = \int_0^{\infty} e^{-st} dt = \left| \frac{e^{-st}}{-s} \right|_0^{\infty}$$

$$f(s) = \frac{1}{s}$$

② Delayed step fun $kU(t-a)$:-

$$f(t) = kU(t-a)$$

$$f(s) = \int_0^{\infty} e^{-st} \cdot kU(t-a) dt = \int_a^{\infty} e^{-st} \cdot k \cdot dt = \frac{ke^{-st}}{s} \Big|_a^{\infty}$$

$$f(s) = \frac{ke^{-as}}{s}$$

③ Ramp fun $k\tau(t)$ or $ktU(t)$:-

$$f(t) = k \cdot \tau(t) = ktU(t)$$

$$f(s) = \int_0^{\infty} ~~kt~~ ktU(t) e^{-st} dt = \int_0^{\infty} k \cdot t e^{-st} dt = k \left[\frac{t e^{-st}}{-s} \right]_0^{\infty} - \int_0^{\infty} \frac{1 \cdot e^{-st}}{-s} dt$$
$$= k(0-0) + \frac{k}{s} \int_0^{\infty} e^{-st} dt = -\frac{k}{s^2} e^{-st} \Big|_0^{\infty}$$

$$f(s) = \frac{k}{s^2}$$

④ Delayed unit ramp fun $r(t-a) :-$

(3)

$$f(t) = r(t-a) = (t-a) U(t-a)$$

$$f(s) = \int_a^{\infty} r(t-a) e^{-st} dt$$

$$= \int_a^{\infty} (t-a) U(t-a) e^{-st} dt$$

$$f(s) = \frac{e^{-as}}{s^2}$$

⑤ Unit Impulse fun $:- \delta(t)$

$$f(s) = \delta(t)$$

$$f(s) = \int_0^{\infty} \delta(t) e^{-st} dt$$

$$f(s) = e^{-st} \Big|_{t=0}^{\infty} = 1 \quad \text{Since } \delta(t) = 1 \text{ only at } t=0$$

$$f(s) = 1$$

⑥ Unit doublet fun $\delta'(t) :-$

$$f(t) = \delta'(t)$$

$$f(s) = s d[\delta(t)]$$

$$= s[1]$$

$$f(s) = s$$

① Step fun:- $f_s(t)$

$$\mathcal{L}[\text{unit step fun}] = \frac{1}{s}$$

$$\mathcal{L}[\text{delayed unit step fun}] = \frac{e^{-as}}{s}$$

$$\mathcal{L}[k U(t)] = \frac{k}{s}$$

$$\mathcal{L}[k U(t-a)] = \frac{k e^{-as}}{s}$$

② Ramp fun:- $f_r(t)$

$$\mathcal{L}[\text{unit ramp fun}] = \frac{1}{s^2}$$

$$\mathcal{L}[\text{Ramp}] = \frac{k}{s^2}$$

$$\mathcal{L}[\text{delayed unit ramp}] = \frac{e^{-as}}{s^2}$$

$$\mathcal{L}[\text{delayed Ramp}] = \frac{k e^{-as}}{s^2}$$

~~as for~~

Differentiation of ramp gives step fun.

Ramp:- $f_r(t) = k r(t) = k t U(t)$

$$f_r(t-a) = k r(t-a) = k t U(t-a)$$

Unit Ramp:- $r(t) = t \cdot U(t)$

$$r(t-a) = (t-a) U(t-a)$$

③ Impulse signal:- $f_s(t)$

$$\mathcal{L}[\text{impulse fun}] = 1$$

Derivative of step fun = impulse fun

$$\frac{d}{dt} [f_s(t)] = f_r(t), \quad \int f_r(t) dt = f_s(t)$$

$$\frac{d}{dt} [f_r(t)] = f_s(t), \quad \int f_s(t) dt = f_r(t)$$

Other Basic Signals:-

① Unit doublet signal

② Sinusoidal "

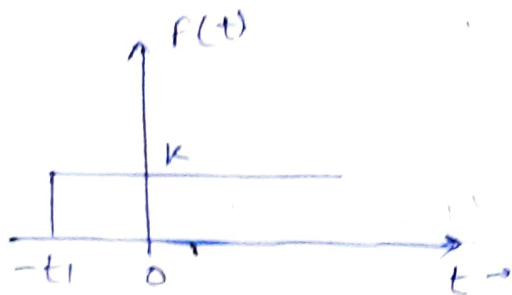
③ Exponential signal

④ Gate signal

eg 1:

Obtain Laplace transform of the waveforms:-

(a)

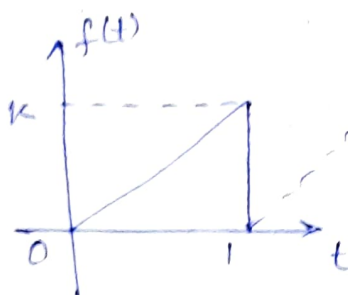


$$f(t) = k U[t - (-t_1)]$$

$$f(t) = k U(t + t_1)$$

$$f(s) = \frac{k e^{t_1 s}}{s}$$

(b)



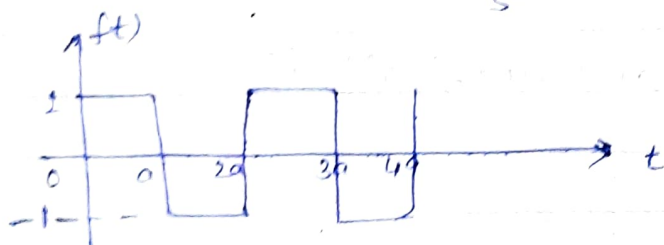
$$f(t) = k t [U(t) - U(t-1)]$$

$$f(s) = \frac{k}{s^2} - \frac{k e^{-s}}{s^2} - \frac{k e^{-s}}{s} = \frac{k}{s^2} (1 - e^{-s} - s e^{-s})$$

$$\mathcal{L}[k t U(t)] = \frac{k}{s^2} \left[\frac{1}{s} - \frac{e^{-as}}{s} \right]$$

$$= \frac{k}{s^2}$$

(c)



$$f(t) = U(t) - 2U(t-a) + 2U(t-2a) - 2U(t-3a) + \dots$$

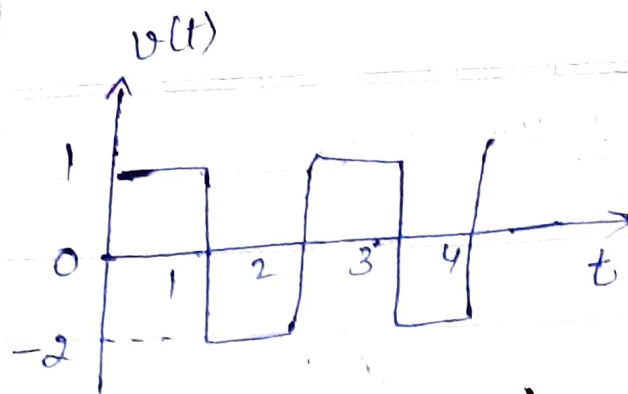
$$f(s) = \left[\frac{1}{s} - \frac{2e^{-as}}{s} + \frac{2e^{-2as}}{s} - \frac{2e^{-3as}}{s} + \dots \right]$$

$$= \frac{1}{s} \left[1 - 2e^{-as} (1 - e^{-as} + e^{-2as} - \dots) \right] \quad \text{G.P}$$

$$= \frac{1}{s} \left[\frac{1 - 2e^{-as}}{1 + e^{-as}} \right] = \frac{1 - e^{-as}}{s(1 + e^{-as})}$$

$$= \frac{e^{as/2} - e^{-as/2}}{s(e^{as/2} + e^{-as/2})} \quad \underline{\text{Ans}}$$

d)



$$v(t) = u(t) - 3u(t-1) + 3u(t-2) - 3u(t-3) + 3u(t-4) + \dots$$

$$= \frac{1}{s} \left[1 - 3e^{-s} + 3e^{-2s} - 3e^{-3s} + 3e^{-4s} + \dots \right]$$

$$= \frac{1}{s} \left[1 - 3e^{-s} (1 + e^{-s} - e^{-2s} + e^{-3s} + \dots) \right]$$

$$= \frac{1}{s} \left[1 - 3e^{-s} (1 + e^{-s})^{-1} \right]$$

$$= \frac{1}{s} \left[\frac{1 - 3e^{-s}}{1 + e^{-s}} \right]$$

$$= \frac{1}{s} \cdot \frac{1 - 2e^{-s}}{1 + e^{-s}} \quad \underline{\underline{\text{Ans}}}$$

① Step function:- $f_s(t)$

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